
UNIVERSITI MALAYSIA PERLIS

Peperiksaan Akhir Semester Pertama
Sidang Akademik 2024/2025

Februari 2025

EMJ12103 – Electric Circuit I
[Litar Elektrik I]

Masa : 3 jam

Please make sure that this question paper has **EIGHT (8)** printed pages including this front page and appendix before you start the examinations.

*[Sila pastikan kertas soalan ini mengandungi **LAPAN (8)** muka surat yang bercetak termasuk muka hadapan dan lampiran sebelum anda memulakan peperiksaan ini.]*

This question paper has **FIVE (5)** questions. Answer **ALL** questions. Each question contributes 20 marks.

*[Kertas soalan ini mengandungi **LIMA (5)** soalan. Jawab **SEMUA** soalan. Markah bagi tiap-tiap soalan adalah 20 markah.]*

Question 1*[Soalan 1]*

- (a) Based on the circuit shown in **Figure 1(a)**, apply **reduce and return approach** to determine the value of the I_Z , V_P , V_X and **power at $5\ \Omega$** .

*[Berdasarkan litar dalam **Rajah 1(a)**, aplikasikan **pendekatan kurangkan dan kembalikan** untuk menentukan nilai I_Z , V_P , V_O dan **kuasa pada $5\ \Omega$** .]*

(10 Marks / Markah)

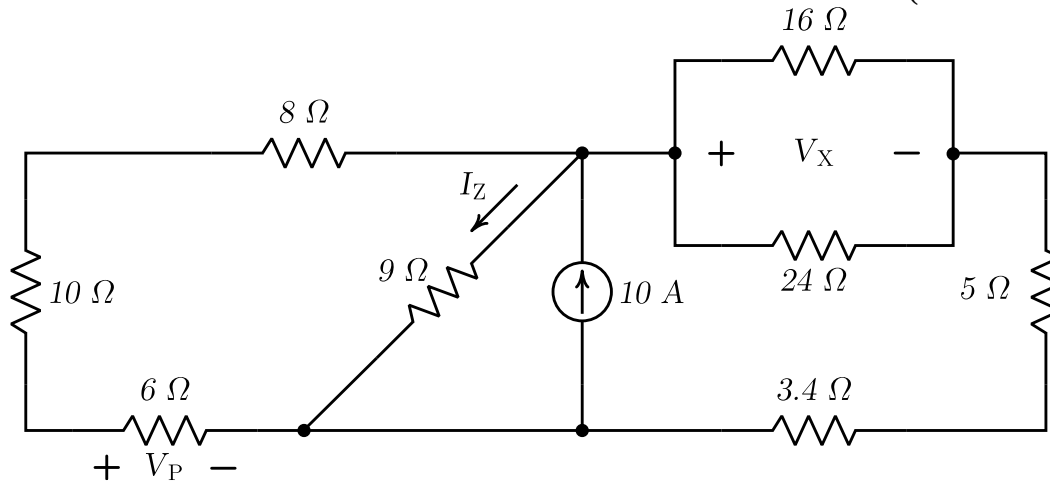


Figure 1(a)
[Rajah 1(a)]

- (b) By applying **delta-star conversion technique**, determine the R_{eq} , V_X and V_Y for the circuit shown in **Figure 1(b)**.

*[Dengan mengaplikasikan **teknik penukaran delta-bintang**, tentukan R_{eq} , V_X dan V_Y bagi litar yang ditunjukkan dalam **Rajah 1(b)**.]*

(10 Marks / Markah)

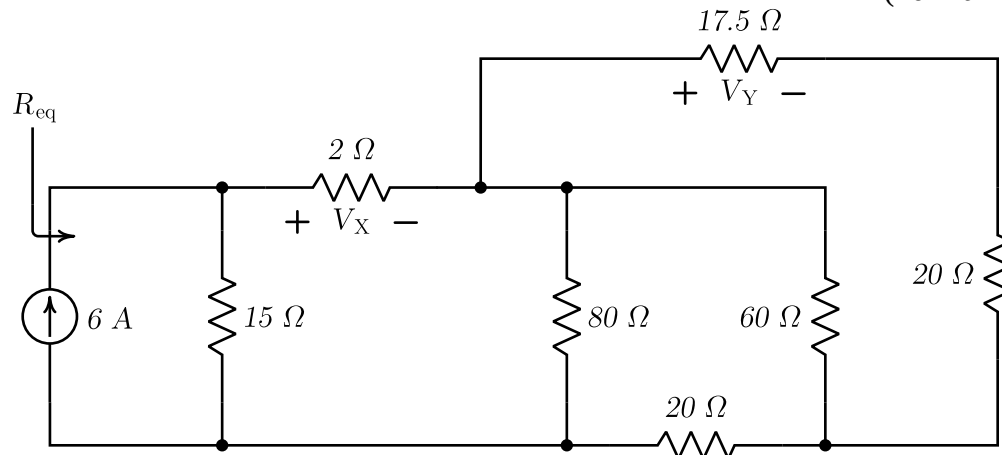


Figure 1(b)
[Rajah 1(b)]

....3/-

Question 2*[Soalan 2]*

- (a) Analyse the circuit in **Figure 2(a)**, determine I_X using **nodal analysis**.

*[Analisa litar dalam **Rajah 2(a)**, tentukan I_X menggunakan **analisa nod.**]*

(10 Marks / Markah)

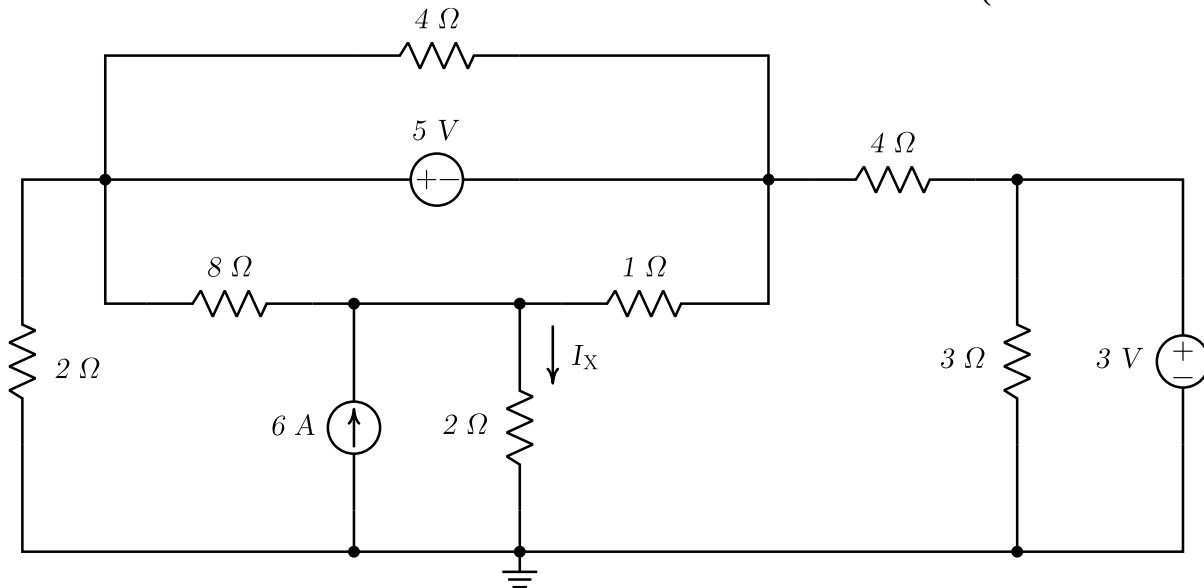


Figure 2(a)
[Rajah 2(a)]

- (b) Analyse the circuit in **Figure 2(b)**, determine I_Y using **mesh analysis**.

*[Analisa litar dalam **Rajah 2(b)**, tentukan I_Y menggunakan **analisa jejaring.**]*

(10 Marks / Markah)

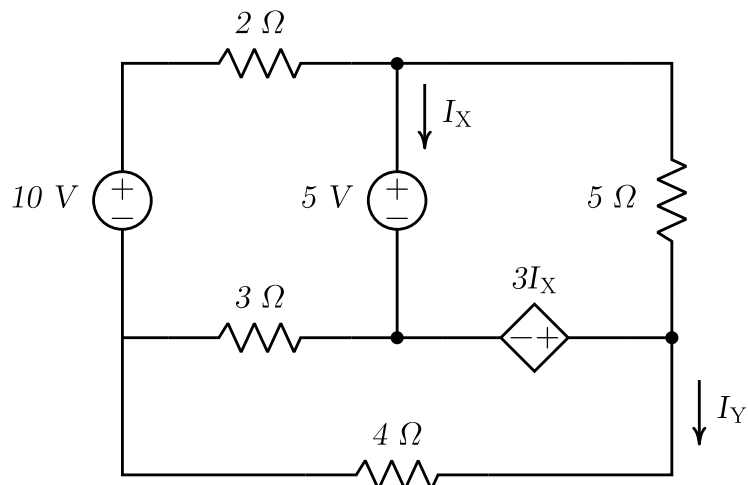


Figure 2(b)
[Rajah 2(b)]

....4/-

Question 3*[Soalan 3]*

- (a) Analyse the value of V_X in **Figure 3(a)** using **source transformation technique**.

*[Analisa nilai V_X dalam **Rajah 3(a)** menggunakan **teknik penjelmaan sumber**.]*

(10 Marks / Markah)

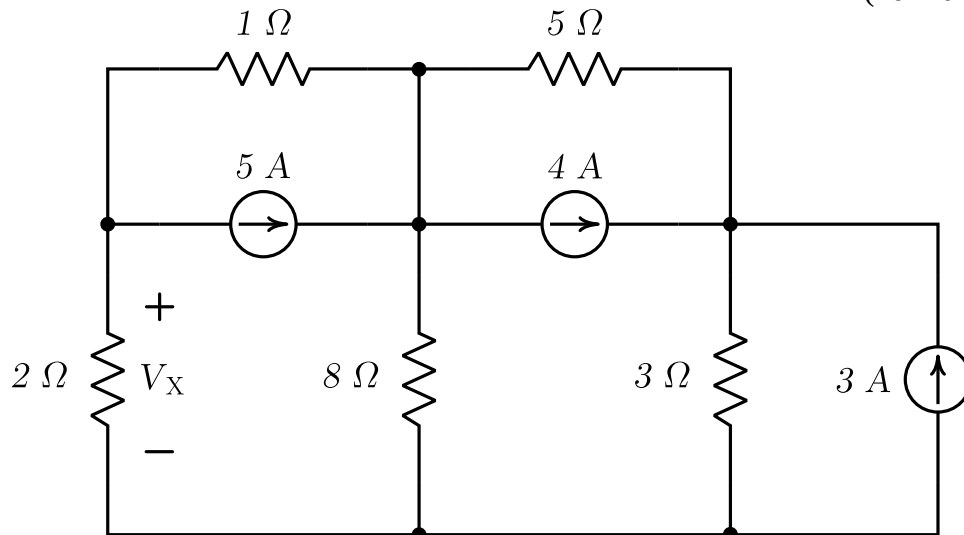


Figure 3(a)
[Rajah 3(a)]

- (b) Based on the circuit in **Figure 3(b)**, analyse V_L by using **superposition technique**.

*[Berdasarkan litar dalam **Rajah 3(b)**, analisa V_L dalam menggunakan **teknik tindihan**.]*

(10 Marks / Markah)

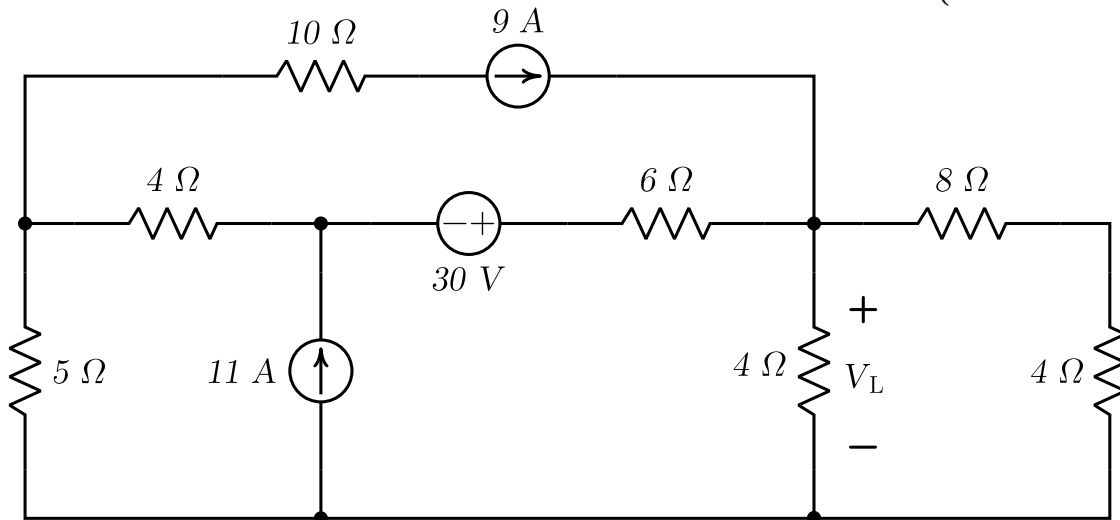


Figure 3(b)
[Rajah 3(b)]

....5/-

Question 4*[Soalan 4]*

- (a) Based on the circuit in **Figure 4(a)**, produce the **Thevenin equivalent circuit** at terminal a-b. Then, calculate the **voltage at $5\ \Omega$** and **maximum power** can be transferred.
*[Berdasarkan litar dalam **Rajah 4(a)**, hasilkan **litar setara Thevenin** pada terminal a-b. Kemudian, kirakan voltan pada $5\ \Omega$ dan kuasa maksimum yang boleh dipindahkan.]*

(10 Marks / Markah)

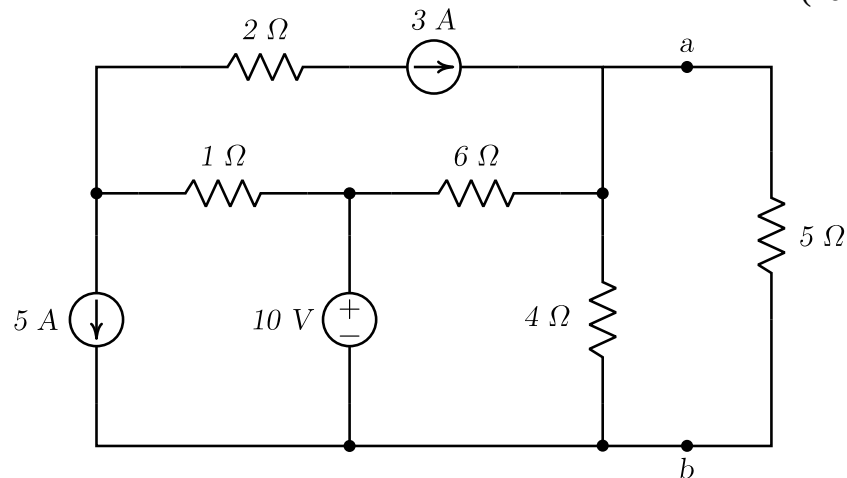


Figure 4(a)
[Rajah 4(a)]

- (b) Based on the circuit in **Figure 4(b)**, produce the **Norton equivalent circuit** at terminal a-b.
*[Berdasarkan litar dalam **Rajah 4(b)**, hasilkan **litar setara Norton** pada terminal a-b.]*

(10 Marks / Markah)

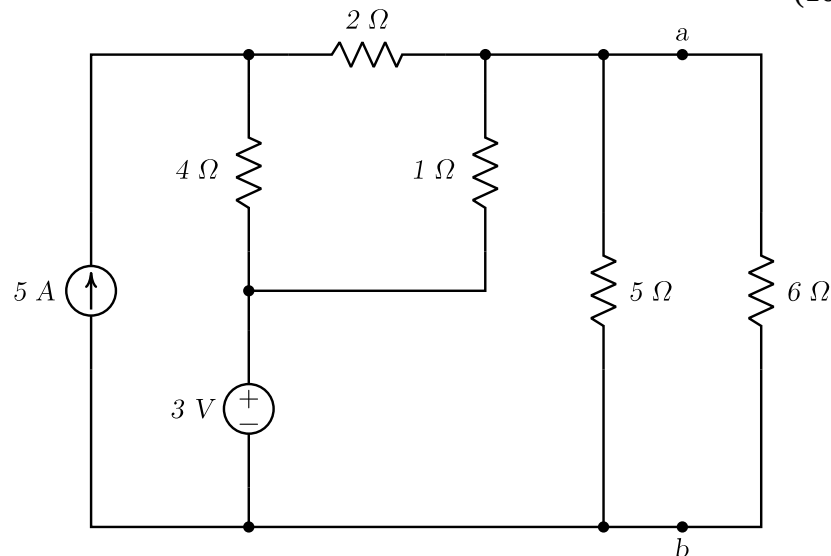


Figure 4(b)
[Rajah 4(b)]

....6/-

Question 5*[Soalan 5]*

- (a) Based on the circuit in **Figure 5(a)**, determine the value of I_L , V_{C1} , V_{C2} and V_{C3} under **DC condition**. Then calculate **energy stored** in each inductor and capacitor.

*[Berdasarkan litar dalam **Rajah 5(a)**, tentukan nilai I_L , V_{C1} , V_{C2} dan V_{C3} dalam **keadaan AT**. Kemudian kirakan **tenaga yang disimpan** dalam setiap pengaruh dan pemuat.]*

(10 Marks / Markah)

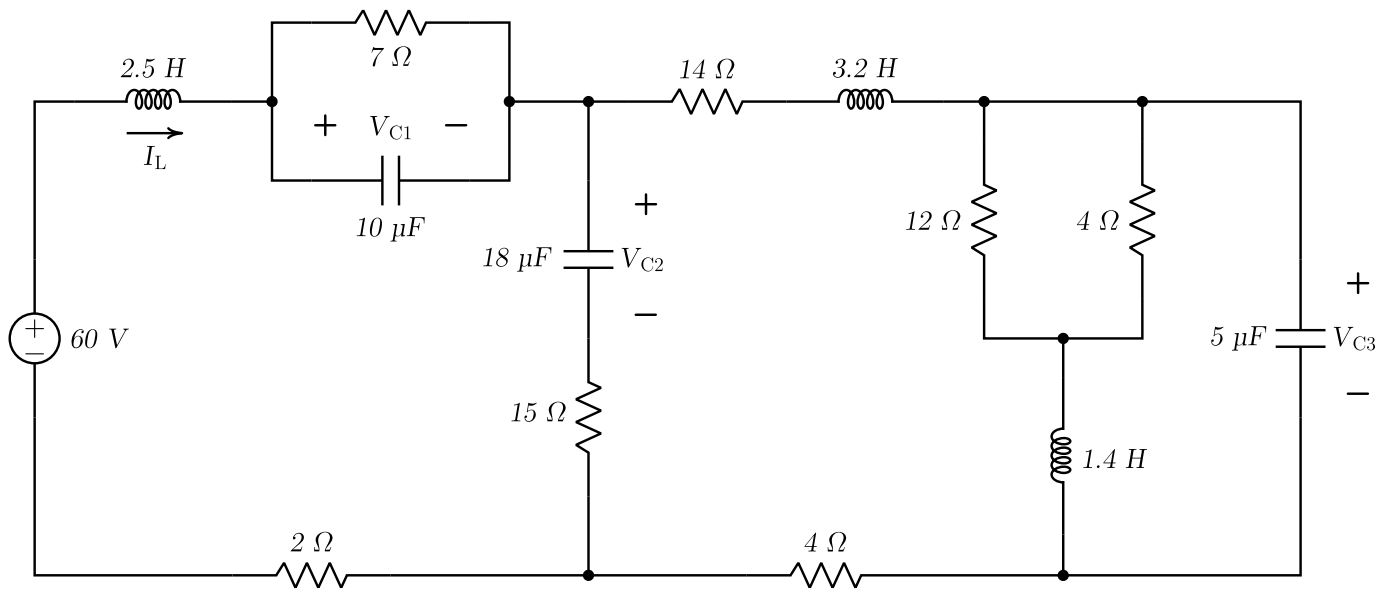


Figure 5(a)
[Rajah 5(a)]

....7/-

- (b) For the circuit in **Figure 5(b)**, $i_s(t) = 6e^{-2t} \text{ A}$. Determine the L_{eq} , $i_1(t)$, $i_2(t)$, $v_1(t)$ and $v_2(t)$.

[Untuk litar dalam **Rajah 5(b)**, $i_s(t) = 6e^{-2t} \text{ A}$. Tentukan L_{eq} , $i_1(t)$, $i_2(t)$, $v_1(t)$ dan $v_2(t)$.]

(10 Marks / Markah)

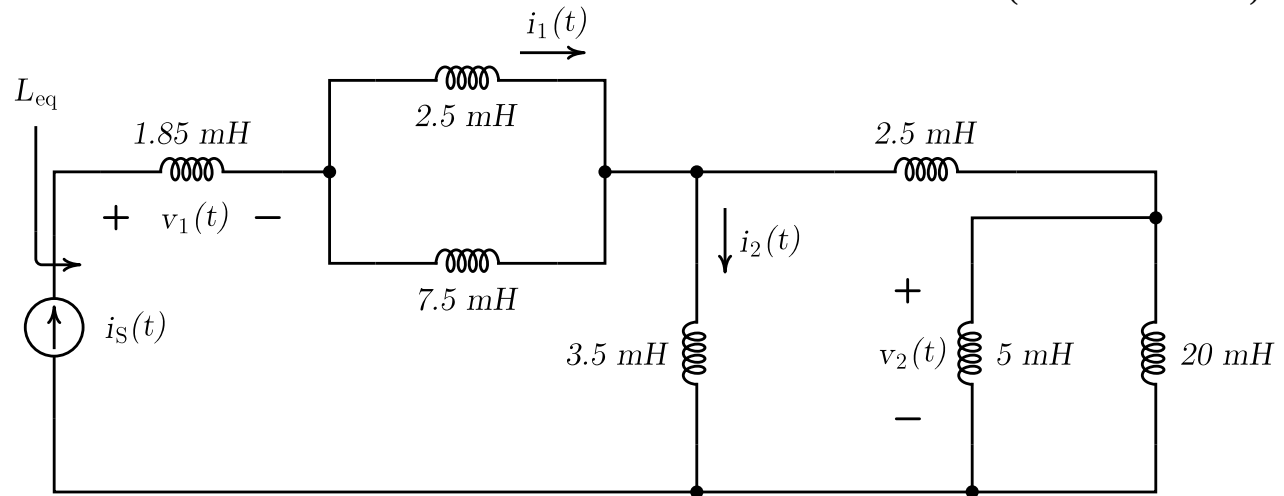
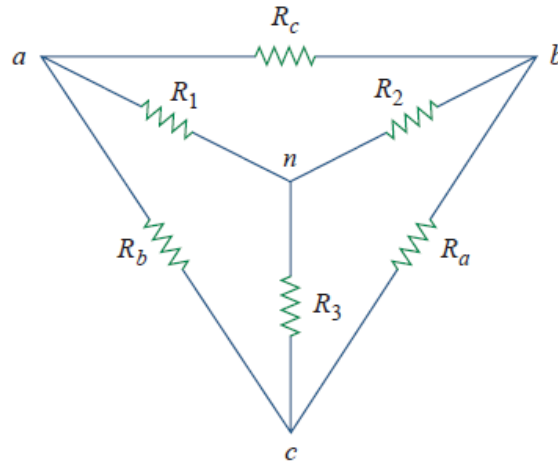


Figure 5(b)
[Rajah 5(b)]

Appendix I

Δ - Y transformation

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

Y - Δ transformation

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_2}$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_3}$$